

Biomechanics of Point-to-Point Reaching

Motor Control Principles for Computational Neuroscience

Week 1 Lecture — Machine Learning for Neuroscience

Graduate Course | Python-Based Curriculum

Lecture Outline

1. Anatomy of Single-Joint Reaching
2. The Triphasic EMG Pattern
3. From EMG to Movement: The Computational Pipeline
4. Equations of Motion and Euler Integration
5. The Bell-Shaped Velocity Profile
6. Antagonist Timing and the Role of the Cerebellum
7. Speed-Accuracy Tradeoff and Fitts' Law
8. Clinical Connections: Dysmetria and Motor Disorders

1. Anatomy of Single-Joint Reaching

The simplest model of a reaching movement involves a single joint (the elbow) rotating the forearm along a **radial arc**. This captures the essential features of voluntary movement while keeping the mechanics tractable.

In our model, the forearm pivots about the elbow from **Point A** (45° flexion) to **Point B** (90° extension), sweeping a 45° arc. The hand traces a circular path at a fixed distance (forearm length $L = 0.35$ m) from the joint center.

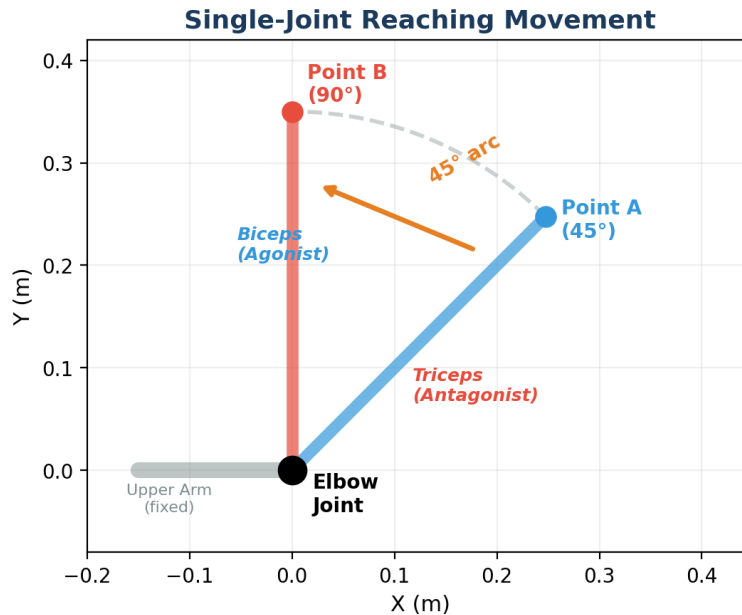


Figure 1: Single-joint reaching geometry. The forearm rotates from 45° to 90° about the elbow.

Physical Parameters

Parameter	Symbol	Value	Units
Forearm length	L	0.35	m
Forearm mass	m	1.5	kg
Moment of inertia	$I = \blacksquare mL^2$	0.0613	$\text{kg}\cdot\text{m}^2$
Joint damping	B	0.5	$\text{N}\cdot\text{m}\cdot\text{s}/\text{rad}$
Start angle	θ_A	45	degrees
Target angle	θ_B	90	degrees

The **moment of inertia** ($I = \blacksquare mL^2$) models the forearm as a uniform rod pivoting at one end. The **viscous damping coefficient** (B) represents passive resistance from joint capsule, tendons, and surrounding tissue.

→ **Key Concept:** The moment of inertia determines how much torque is needed to produce a given angular acceleration. Larger I means the limb resists changes in motion more strongly.

2. The Triphasic EMG Pattern

Rapid, goal-directed movements are not produced by simply turning on the agonist muscle. Instead, the central nervous system generates a precisely timed sequence of muscle activations known as the **triphasic EMG pattern** (Hallett et al., 1975; Wacholder & Altenburger, 1926).

Burst	Muscle	Timing	Function
AG1	Agonist (Biceps)	~50–150 ms	Accelerate limb toward target
ANT	Antagonist (Triceps)	~120–260 ms	Decelerate and brake the limb
AG2	Agonist (Biceps)	~250–350 ms	Dampen oscillation, stabilize at target

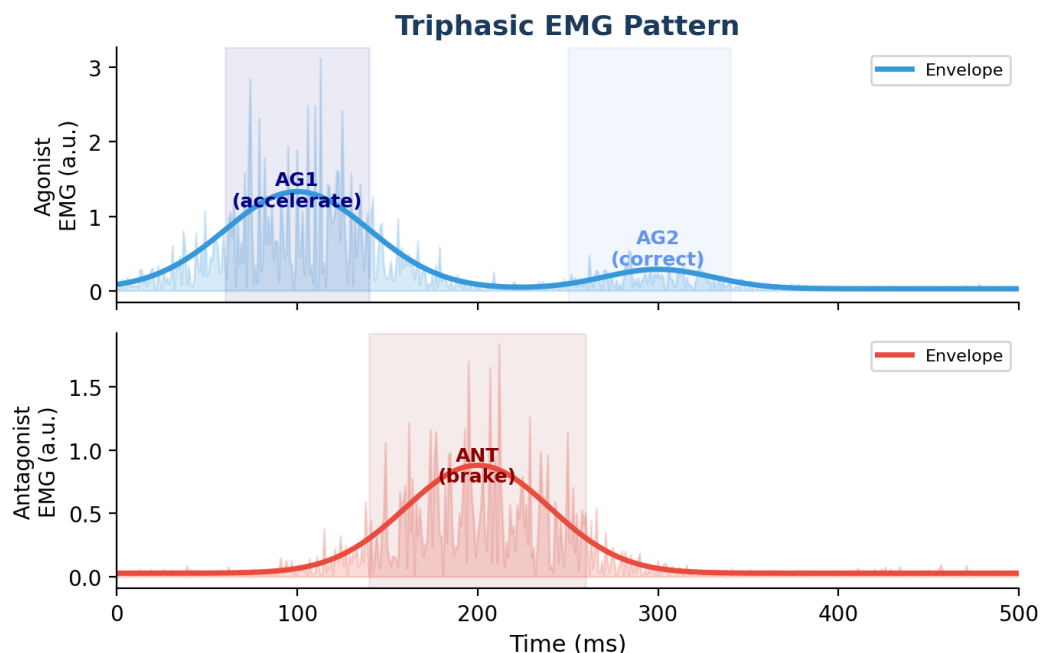


Figure 2: The triphasic EMG pattern. AG1 accelerates the limb, ANT brakes it, and AG2 corrects residual error.

EMG Signal Model

We model each EMG burst as a Gaussian-shaped **envelope** (representing smooth motor unit recruitment and de-recruitment) multiplied by **rectified noise** (representing stochastic motor unit action potentials):

$$EMG(t) = A \cdot \exp(-t - \mu)^2 / (2\sigma^2)) \cdot |noise(t)|$$

where A is the peak amplitude, μ is the burst center time, and σ controls the burst duration.

→ **Key Concept:** The triphasic pattern is pre-programmed (feedforward), not simply a reflex. It appears even in deafferented patients who lack sensory feedback, demonstrating that the CNS generates the entire sequence before movement onset.

3. From EMG to Movement: The Computational Pipeline

Converting EMG signals into limb movement requires a three-stage pipeline:

- ① **EMG → Muscle Force:** The rectified, smoothed EMG envelope is approximately proportional to muscle force (linear EMG-force relationship at moderate contraction levels).
- ② **Muscle Forces → Net Joint Torque:** The agonist produces positive torque (toward the target) and the antagonist produces negative torque (opposing motion). The net torque is their weighted sum.
- ③ **Net Torque → Movement (Newton's 2nd Law):** The angular equation of motion governs how torque produces rotation.

Computational Pipeline: EMG → Movement

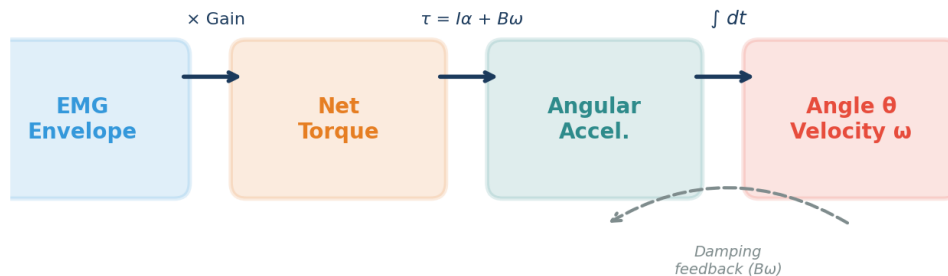


Figure 3: Computational pipeline from EMG activation to joint kinematics.

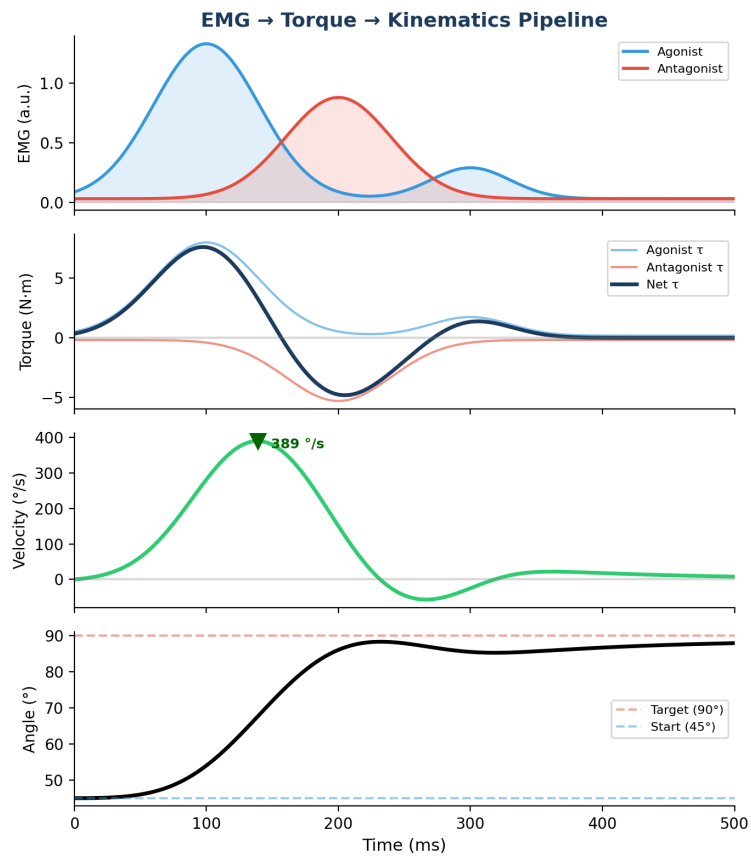


Figure 4: Simulation results. EMG envelopes (top) produce torques that drive a realistic reaching movement.

4. Equations of Motion and Euler Integration

The forearm rotation is governed by Newton's second law for rotation:

$$I \cdot \alpha(t) = \tau_{net}(t) - B \cdot \omega(t)$$

where I is the moment of inertia, $\alpha = d\omega/dt$ is angular acceleration, τ_{net} is the net muscle torque, B is the damping coefficient, and $\omega = d\theta/dt$ is angular velocity.

Rearranging to solve for acceleration:

$$\alpha(t) = [\tau_{net}(t) - B \cdot \omega(t)] / I$$

Euler Integration

We solve this ordinary differential equation numerically using **Euler's method**, the simplest numerical integration scheme. At each time step dt :

$$\omega(t + dt) = \omega(t) + \alpha(t) \cdot dt$$

$$\theta(t + dt) = \theta(t) + \omega(t) \cdot dt$$

Starting from rest ($\omega = 0$) at the initial angle ($\theta = 45^\circ$), we step forward in 1 ms increments, updating velocity and position at each step. Despite its simplicity, Euler integration is accurate enough for our 1 ms time steps (the dynamics change slowly relative to dt).

➔ **Key Concept:** The damping term ($B \cdot \omega$) acts like friction — it opposes motion proportional to velocity. This is why the limb eventually stops even without a perfect braking burst: the damping passively dissipates kinetic energy.

The Role of Torque Gain

The **torque gain** (set to 6.0 N·m per unit EMG in our simulation) converts the dimensionless EMG envelope into physical torque. This gain encapsulates the muscle's force-generating capacity, moment arm, and the number of recruited motor units. With our tuned parameters, the simulated movement reaches 88.9° (target: 90°), producing a realistic 1.1° endpoint error.

5. The Bell-Shaped Velocity Profile

One of the most robust findings in motor control is that point-to-point reaching movements produce a **bell-shaped (single-peaked) velocity profile**. The limb accelerates smoothly to a peak velocity near the midpoint of the movement, then decelerates symmetrically.

This characteristic shape emerges from the **minimum jerk model** (Flash & Hogan, 1985), which proposes that the CNS plans movements to minimize the integral of squared jerk (the rate of change of acceleration). The resulting optimal trajectory predicts:

- A symmetric, bell-shaped velocity profile with peak at $t = 0.5 \times$ movement time
- Smooth, continuous acceleration with no abrupt changes
- A **symmetry ratio** of exactly 0.5 (time to peak velocity / total movement time)

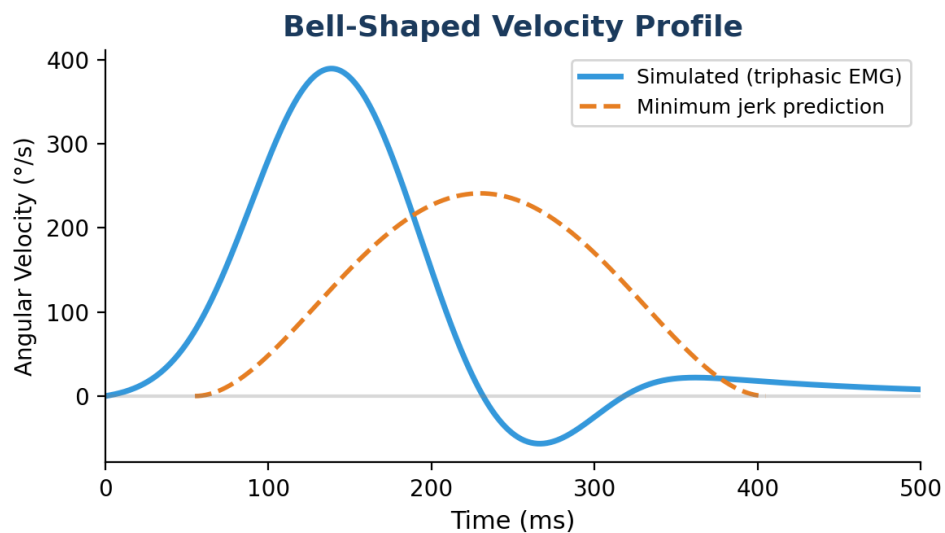


Figure 5: The simulated velocity profile (blue) compared to the minimum jerk prediction (orange dashed).

Our triphasic EMG model naturally produces an approximately bell-shaped profile, validating that the AG1-ANT-AG2 timing captures the essential control strategy. Deviations from perfect symmetry arise because the EMG model is not explicitly optimizing jerk.

→ **Key Concept:** The symmetry ratio quantifies movement quality. Values near 0.5 indicate smooth, well-planned movements. Ratios below 0.4 suggest the deceleration phase is prolonged (common in pathological movements).

6. Antagonist Timing and the Role of the Cerebellum

The precise timing of the antagonist (braking) burst is the single most critical parameter for movement accuracy. Our simulations reveal dramatic effects when the ANT onset is shifted by as little as 40–80 ms:

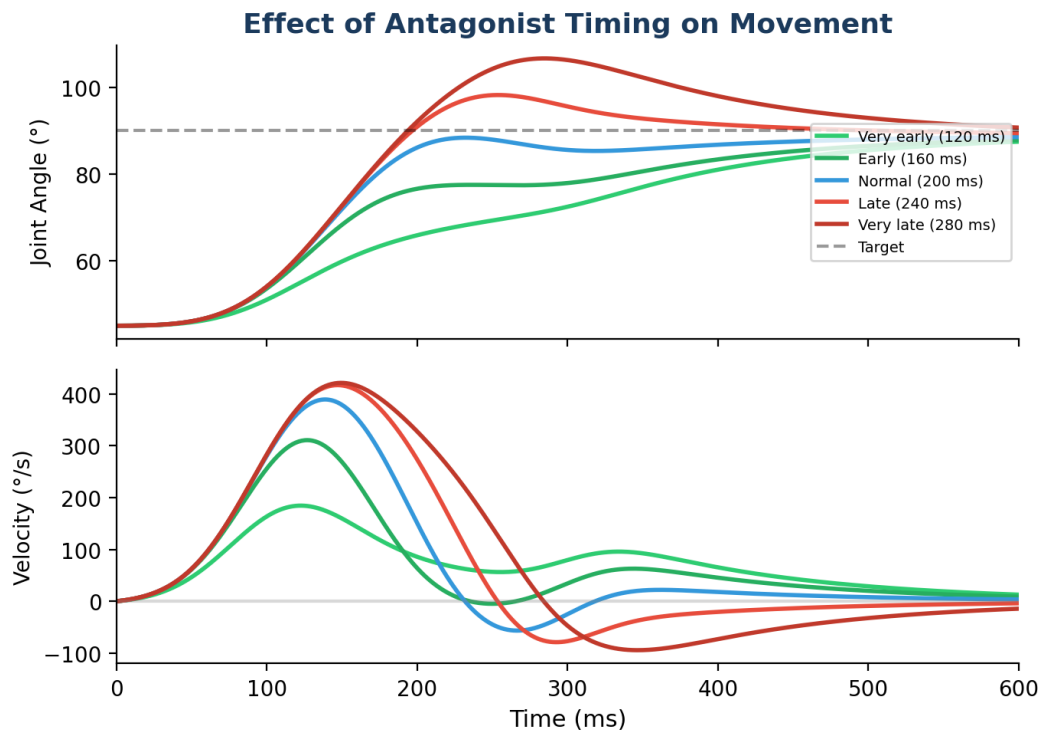


Figure 6: Effect of antagonist timing. Late braking produces higher peak velocity and significant overshoot.

ANT Timing	Peak Velocity	Overshoot	Movement Character
Very early (120 ms)	185 °/s	0.0°	Slow, cautious approach
Early (160 ms)	311 °/s	0.0°	Moderate, smooth
Normal (200 ms)	389 °/s	0.0°	Fast, accurate
Late (240 ms)	417 °/s	9.3°	Overshoots, corrects
Very late (280 ms)	422 °/s	17.7°	Large overshoot, oscillation

The Cerebellum as a Timing Engine

The **cerebellum** is critical for predicting the correct ANT onset time. It acts as a forward model, estimating the current limb state and computing when braking torque must begin to stop at the target. This prediction must account for sensorimotor delays (~100–200 ms), meaning the antagonist command is largely **pre-programmed** before proprioceptive feedback about the movement is available.

→ Key Concept: A 120 ms shift in ANT timing doubles the peak velocity (185 → 422 °/s) and introduces 17.7° of overshoot. This explains why cerebellar patients exhibit dysmetria — they cannot accurately time the braking burst.

7. Speed-Accuracy Tradeoff and Fitts' Law

A fundamental principle of motor control is the **speed-accuracy tradeoff**: faster movements are less accurate. Paul Fitts formalized this in 1954 with what is now known as **Fitts' Law**:

$$MT = a + b \cdot \log_2(2D / W)$$

where MT is movement time, D is the distance to the target, W is the target width, and a, b are empirical constants. The term $\log_2(2D/W)$ is the **index of difficulty**.

In our EMG framework, faster movements require:

- **Higher AG1 amplitude** (more motor unit recruitment = more force)
- **Narrower burst widths** (shorter activation duration)
- **Compressed inter-burst timing** (ANT must follow AG1 more quickly)
- **Proportionally larger ANT burst** (more braking force needed)

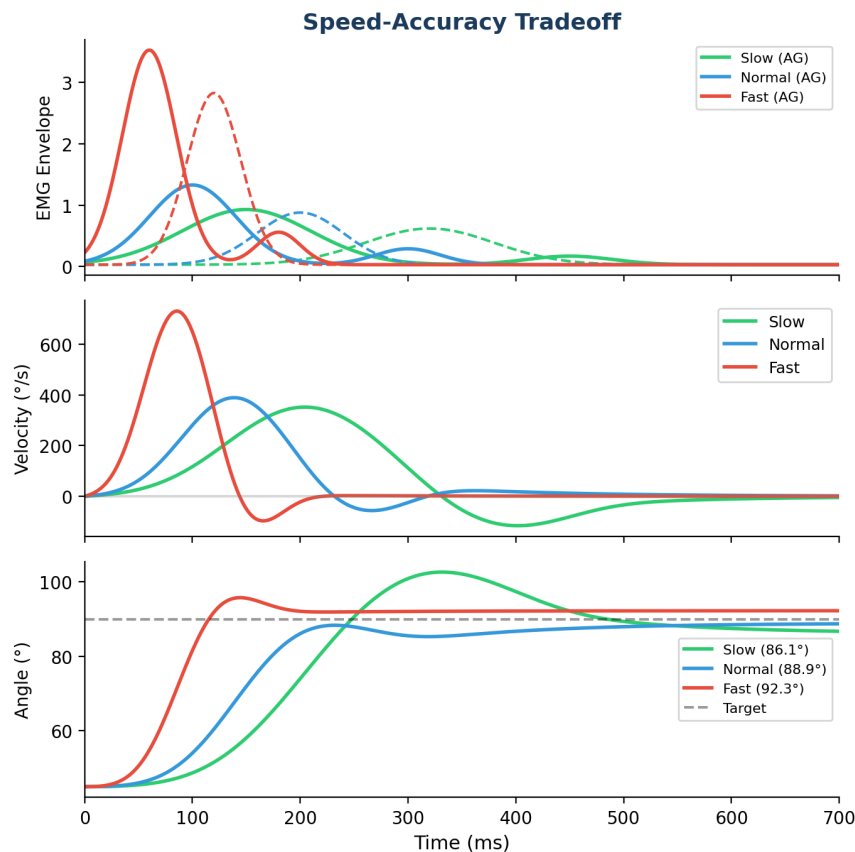


Figure 7: Speed-accuracy tradeoff. Fast movements reach quickly but overshoot; slow movements are more accurate.

Condition	Movement Time	Endpoint Error	Peak Velocity
Slow	679 ms	3.9°	352 °/s
Normal	543 ms	1.1°	389 °/s
Fast	209 ms	2.3°	732 °/s

The origin of the speed-accuracy tradeoff lies in **signal-dependent noise**: larger neural commands have proportionally more variability, meaning the motor system cannot simply scale up force without also scaling up the uncertainty in force output (Harris & Wolpert, 1998).

8. Clinical Connections

The biomechanical principles in this lecture map directly to clinical motor disorders. Understanding the triphasic EMG model provides a framework for interpreting pathological movement patterns.

Condition	Disrupted Component	Movement Signature
Cerebellar ataxia (dysmetria)	ANT burst timing	Overshoot, oscillation, multiple velocity peaks
Parkinson's disease (bradykinesia)	AG1 burst amplitude	Slow movements, reduced peak velocity, undershooting
Huntington's disease (chorea)	Burst timing variability	Involuntary movements, irregular trajectories
Spasticity (upper motor neuron)	Tonic antagonist activity	Increased joint resistance, co-contraction
Essential tremor	AG/ANT co-oscillation	Rhythmic oscillation during movement

Connecting to Machine Learning

In later weeks of this course, we will use machine learning to:

- **Decode movement intention** from EMG signals using SVMs and neural networks
- **Classify movement disorders** from kinematic features using random forests and LDA
- **Predict motor outcomes** from brain imaging data using GLMs
- **Cluster movement patterns** to discover motor subtypes using k-means

The biomechanical model from this week provides the domain knowledge needed to engineer meaningful features and interpret ML results in a neuroscience context.

➔ **Today's Lab: You will implement this entire pipeline in Python — generating triphasic EMG signals, simulating joint dynamics, and analyzing how parameter changes affect movement quality. See the lab notebook for hands-on exercises.**

Key References

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